

Zain Dana Harper

Curriculum vitae. Independent systems and verification researcher, Seattle.

Seattle, Washington | zaindharper@gmail.com | harperz9.github.io | github.com/HarperZ9 (<https://github.com/HarperZ9>) |
ORCID 0009-0001-7175-5393 (<https://orcid.org/0009-0001-7175-5393>)

Profile

My work is about seams: the places where evidence is lost between systems, where a claim picks up more meaning than its observation supports, and where a human decision has to stay explicit. The engineering and the research are the same question asked twice.

Public engineering practice since 2023, on roughly thirteen years of programming and eleven years running tree crews from the ground. The route was not academic and the record says so plainly. What that route produced is an attention to handoffs. A decision can be locally correct and globally wrong because an observation was dropped in transit, a changed condition went unrecorded, or the person with authority received a compressed version of the evidence.

Current lanes are verification infrastructure for AI agents, compilers and typed effects, evaluation provenance, real-time graphics and color, and the philosophy of made minds. The long-range destination is frontier AI, epistemics, and the cognitive science next to them, with robotics as the point where these questions stop being about text.

Research agenda

Witnessed Systems and Bounded Agency

Made systems may be granted operational standing, and their claims and actions should stay bounded by evidence that can be independently re-derived, evaluation independence that is recorded rather than assumed, and explicit human authorization at consequential seams. Five workstreams carry it. The full program, its twelve-month plan, and what it does not claim are in the dossier.

- **Evaluation provenance and independent witness.** A machine-readable receipt recording evaluator identity, model and dataset versions, shared dependencies, negative controls, and whether the verifier inherited claims from the system it assessed.
 - **Live-state verification and drift.** Separating identity, integrity, behavior, and authorization for an agent or controller that may drift between evaluation and deployment, with MATCH, DRIFT, and UNVERIFIABLE as real outcomes instead of missing evidence laundered into confidence.
 - **Proof-carrying agent actions.** A minimal envelope for a consequential action, tested on whether it survives tool adapters and multi-agent handoffs without changing meaning.
 - **Epistemic and cognitive dynamics.** The gap between what an agent represents, says, and does, separating uncertainty from confabulation, memory failure, and policy conflict. Human parallels stay analogies and never claims of equivalence.
 - **Embodied and robotic authorization.** What evidence a robot needs before crossing a safety boundary, how sensor uncertainty should propagate into action permissions, and how a witness stays advisory instead of quietly becoming a second controller.
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Publications

Eight public records with permanent DOIs. Status is stated exactly, and none of this is peer reviewed: it is systems papers, published preprints, research notes, and archived corpora. Where a venue or a documented independent review process exists, this page will name it.

EMET: An Authority-Incapable Byte-Level Integrity Witness

SYSTEMS PAPER · 2026

Re-derives byte-integrity facts and refuses to emit permission, trust, safety, or authority. Six boundaries form one perimeter separating an is-question from an ought-question. Public implementation in four independent versions.

doi.org/10.5281/zenodo.21230267 (<https://doi.org/10.5281/zenodo.21230267>)

BuildLang: Accountable Compute via Typed Capability Effects

SYSTEMS PAPER · 2026

Ambient authority made part of a function's type, with a public compiler behind it. doi.org/10.5281/zenodo.21231253 (<https://doi.org/10.5281/zenodo.21231253>)

Witnessed Independence: Recording Whether a Verifier Graded Its Own Work

PUBLISHED PREPRINT · 2026

doi.org/10.5281/zenodo.21232206 (<https://doi.org/10.5281/zenodo.21232206>)

Proof Packets: A Derive-Don't-Trust Envelope for Accountable Agent Actions

PUBLISHED PREPRINT · 2026

doi.org/10.5281/zenodo.21231406 (<https://doi.org/10.5281/zenodo.21231406>)

The Personhood-Gate Handoff

RESEARCH NOTE · 2026

doi.org/10.5281/zenodo.21234475 (<https://doi.org/10.5281/zenodo.21234475>)

Re-Perceived Effects

RESEARCH NOTE · 2026

doi.org/10.5281/zenodo.21231311 (<https://doi.org/10.5281/zenodo.21231311>)

The Witnessing Spine

ARCHIVED RESEARCH CORPUS · 2026

doi.org/10.5281/zenodo.20778927 (<https://doi.org/10.5281/zenodo.20778927>)

Conferred Existence

THESIS, ARCHIVED CORPUS · 2026

Long-form philosophical work on made minds, perception, memory, external anchors, legitimate standing, and relation to the world. It supplies the ground the verification work stands on. doi.org/10.5281/zenodo.20773724 (<https://doi.org/10.5281/zenodo.20773724>)

The 2026 verification engine

Flywheel

Accountability engine and native client. Python stdlib-only engine, Flutter client. On the public default branch as of 2026-08-03.

- A verified-inference loop where a model acts only through a default-deny capability gate it cannot talk past. Writes, exec, and external tools stay off until a human grants them, and the model cannot self-authorize.
- Every tool invocation emits a sealed record binding the capability class, the admission decision, and witnessed argument and output digests, never raw content. Records are hash-chained, signed, and offline-verifiable, and verification recomputes every seal from the record alone.
- A transitive witness graph degrades exactly the dependents of a drifted action, alongside an append-only learning loop turning witnessed divergences into hash-bound, re-checkable lessons.
- An infrastructure control package: network egress receipts, credential scanning, isolation acceptance tests, a dual-confirmation kill switch that is off by default with cloud credential revocation still a stub, and cross-layer event correlation that is heuristic today.
- Capability-typed shell admission with a Unicode-spoof neutralizer, a domain oracle registry across three live domains, a model-neutral router with quota failover, a subscription-auth adapter that consumes an authorized token rather than minting one, and session tooling to browse, resume, and export past verified runs.

The trained local model is the replaceable half of the loop and the engine is the durable half. No capability uplift is claimed for the model. The first preregistered confirmatory run completed 2026-08-04. Across 2,646 certificate bodies through the accept path in nine model contexts there were zero verdict disagreements, so acceptance is provably a function of the certificate. Oracle selection beat random selection at every model size on the solvable task family, scored by an independently written held-out checker with random and placebo controls, paired delta up to +0.32 and exact p to 3.8e-06. The harder family defeated every model on the ladder; that null is kept and labeled uninformative, and the self-scored comparison is refused by design. Run and analysis are anchored in a signed, offline-verifiable transparency ledger.

Public systems

BuildLang

A systems language where ambient access is part of a function's type. Typed capability effects for filesystem, network, foreign calls, and gated compile-time macros, tracked through function values, closures, struct fields, control flow, and async blocks. Hindley-Milner inference, sum types, an opt-in experimental linear-types attribute with honestly scoped soundness, two-way C FFI, a production C backend, HLSL and GLSL shader output, and experimental SPIR-V, LLVM IR, WebAssembly, Rust, x86-64, and ARM64 backends. The buildc CLI bundles build, run, test, repl, fmt, pkg, watch, doctor, and an LSP server.

Project Telos tool family

Index for repository maps, dependency graphs, and context packs. Gather for research intake that carries provenance. Forum for model-agnostic orchestration on a replayable ledger. Crucible for worker and verifier evaluation with MATCH, DRIFT, and UNVERIFIABLE verdicts. Learn for accountable learning. EMET for byte-level witnessing at a frozen 1.0.0 specification. Public source packages for routing, toolchain auto-wiring, and provenance-carrying agent memory. Registry publication is not claimed where a release has not been verified.

AI evaluation environments

A published evaluation environment, mlflow-terminal-state on Prime Intellect's Environments Hub under zaindanaharper, scores an AI coding agent across 324 cases by the terminal state of the environment it worked in rather than by what its transcript claims. Source, pinned-claim tests, and CI are public at terminal-state-fixtures (<https://github.com/HarperZ9/terminal-state-fixtures>).

Graphics, color, and creative tooling

Public HLSL and C++ graphics work including a Skyrim post-processing project past 930,000 downloads, plus D3D11 and HLSL systems for tone mapping, TAA, SSR, SSGI, GTOA, volumetrics, GPU traces, and read-only shared-memory bridges. Color science covers perceptual spaces, color difference, HDR tone mapping, gamut tools, ICC and VCGT workflows, and display characterization. The browser-native side is live on this site: a generative studio, a print desk and exhibition, a retro rendering engine, and a loom that reads and writes WIF and proves its own export.

Upstream open source

Pull requests into repositories owned by other people. Counts come from the GitHub API through tools/pr_census.py, and a test fails the build if this page disagrees with the live record.

49 engineering pull requests

29 merged

9 open

11 closed without merge

Merged work sits in Datasette, tomlkit, pydantic-ai, DeepEval, pydash, grimp, agentfence, and a set of smaller agent and evaluation projects, across 24 repositories. Representative accepted changes: correct HTTP status handling for malformed composite-key row URLs, a TOML round-trip bug where an array of tables swallowed the next sibling key, a metric crash on unhashable tool outputs, missing module and layer validation errors, seeded determinism and mid-task recovery fixtures for agent runtimes, and, in August 2026, a policy-engine allocation reduction and a governance evidence-mapping document in agentfence.

A further 14 pull requests only add a row naming a tool to a public link index. They change no software and are counted apart. Open is not accepted, and the declined work stays visible on the portfolio ledger.

Speaking and service

- **Puget Sound Programming Python.** "Pick the Lock for Everyone: Building Verifiable AI Workflows in Python," invited, 25 minutes, 2026-08-19.
 - **Coalition for Secure AI.** Individual contributor to Workstream 4, Secure Design for Agentic Systems, contributing secure-design patterns and a reference implementation as of August 2026.
 - **PlantAmnesty, Seattle.** Volunteer time each year since 2015 supporting a community network of horticulturists and arborists, its workshops, and its member events. Self-reported; a service-history letter has been requested.
 - **Wildfire relief near Tonasket, Washington, 2016.** Packed, prepared, and delivered essentials for displaced people. Self-reported, pending a corroborating record.
 - **Companionship volunteering, The Springs at Wilsonville.** Regular visits with an elderly resident during high school. Details stay private out of respect for the resident and his family.
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Professional experience

Independent systems engineer and founder · Zentropy Labs

2023 TO NOW · SEATTLE AND REMOTE

- Build and maintain the public systems portfolio across Python, Rust, C++, HLSL, TypeScript and JavaScript, and web-native surfaces.
- Own architecture, implementation, test strategy, documentation, packaging, release notes, demos, public claims, and maintenance planning.
- Use coordinated coding agents for parallel discovery, implementation, and review, and keep first-party architecture, integration, verification, and release decisions explicit.

Freelance technical writer, GRC documentation, product operations

2017 TO NOW · REMOTE

- API and implementation guides, security and compliance documentation, proposals, RFP and grant material, release notes, and operational playbooks.
- NIST 800-171, CMMC readiness, SOC 2, ISO 27001, DFARS, incident response, and audit support, in a technical-writing capacity.
- Client names, private deliverables, compliance evidence, and contract terms stay outside the public portfolio.

Operations and commercial arboriculture lead · family business

2015 TO NOW · SEATTLE AREA

- Eleven years operating tree crews from the ground. Ran the rigging systems, judged clearances and how far limbs would swing relative to structures and people, and served as the second set of eyes for the person in the air.
- Led client intake, estimates, site assessment, scheduling, safety judgment, crew and vendor coordination, proposals, and customer communication.

Technical networking support, Xbox division · Microsoft

2013 TO 2014 · REDMOND, WASHINGTON

- Diagnosed TCP/IP, DNS, NAT, firewall, router, and account-adjacent console networking issues across phone and chat support.
- Documented repeatable resolutions and translated technical fixes into guidance a customer could follow.

Education and self-directed study

- **Wilsonville High School, Wilsonville, Oregon.** Diploma, 2013.
- **Self-directed since.** Roughly thirteen years of programming practice, public and release-driven from 2023. Compilers, type systems, graphics, color science, verification, and the philosophy of mind were learned from primary sources, from source code, and from the consequences of shipping.
- **Applied domain knowledge.** Arboriculture, horticulture, botany, and biological systems from eleven years of field practice, alongside self-directed study of anatomy, physiology, exercise science, and nutrition. This is applied and self-directed knowledge, not licensed expertise or clinical training, and it is never presented as either.

Technical skills

CORE COMPETENCIES	AI agent evaluation and testing, QA and test automation, policy enforcement and audit logging, developer tooling, technical and compliance documentation, release engineering, customer-facing technical support
LANGUAGES	Python, Rust, C++, TypeScript and JavaScript, Lua, HLSL, C#, PowerShell, Bash
VERIFICATION	Capability gates, sealed receipts, hash chains, offline re-verification, witness graphs, provenance, preregistration discipline, evidence contracts, conformance corpora
AI SYSTEMS	Multi-provider and local-endpoint routing, tool-use loops, MCP surfaces, task decomposition, multi-agent orchestration, worker and verifier separation, evaluation workflows
COMPILERS	Lexing, parsing, AST and intermediate representations, type checking, typed capability effects, ownership and lifetime analysis, code generation, native C output, shader output, editor tooling
GRAPHICS	D3D11 and HLSL rendering, proxy-DLL architecture, post-processing, shared-memory IPC, color spaces and appearance models, HDR and tone mapping, ICC, VCGT, and 3D-LUT workflows, display characterization, WebGL and Canvas
DELIVERY	Git and GitHub, pytest, Node test tooling, GitHub Actions, Linux, CMake, vcpkg, package metadata, CLI design, static sites
WRITING	Developer and API documentation, implementation and support guides, proposals, release documentation, compliance and audit-support writing, long-form technical essays

Work preferences

Remote preferred, open to onsite or hybrid, and to contract, full-time, part-time, and project-based engagements. The strongest fits are AI and agent infrastructure, verification and security tooling, compiler and language tooling, graphics and color, research operations, technical writing, and systems work in a domain I have to learn from scratch.

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Updated 2026-08-21. Claim status is stated per item. Nothing here is peer reviewed; where that changes, the entry will name the venue. Open-source counts are read from the GitHub API and gated by a test.